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3 Complex Analysis 2024
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Test 2
 9 October 2024



Instructions:

- This question paper consists of **7 pages** (including this one).
- You have 90 minutes to answer the questions.
- Answer all questions in the spaces provided on this question paper. Use the backs of pages if needed.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 40 points = 100% (42 points available).

Problem	1	2	3	4	Σ	Mark
Maximum Points	12	8	10	12	40	100%
Result						

Student Number:									
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Problem 1

(12 points)

Calculate the following integrals with the help of Cauchy's Integral Theorem, Cauchy's Integral Formula, or Cauchy's Integral Formula for derivatives.

- (a) $\int_{\gamma} \frac{\sin z}{z} dz$, where $\gamma(t) = e^{it}$, $t \in [0, 2\pi]$
- (b) $\int_{\gamma} \frac{z^2}{z^2 - 3z + 2} dz$, where $\gamma(t) = 5e^{it}$, $t \in [0, 2\pi]$
- (c) $\int_{\gamma} \frac{e^z}{(z^2 + 4)^2} dz$, where $\gamma(t) = -i + 2e^{it}$, $t \in [0, 2\pi]$

Solution.

- (a) 1st solution: Since $\lim_{z \rightarrow 0} \frac{\sin z}{z} = 1$, $z_0 = 0$ is a removable singularity of the function $f(z) = \frac{\sin z}{z}$. I.e., by putting $f(0) = 1$, we extend it to a holomorphic function on $\mathbb{C}$. By Cauchy's Integral Theorem, $\int_{\gamma} \frac{\sin z}{z} dz = 0$.

2nd solution: Notice that the point $z_0 = 0$ lies inside the circle γ . By Cauchy's Integral Formula we have

$$\int_{\gamma} \frac{\sin z}{z} dz = 2\pi i \sin 0 = 0.$$

- (b) The zeros of the denominator $z^2 - 3z + 2$ are

$$z_{1,2} = \frac{3}{2} \pm \sqrt{\frac{9}{4} - 2} = \frac{3}{2} \pm \frac{1}{2},$$

i.e. $z_1 = 1$, $z_2 = 2$. Both of them lie inside γ .

We determine the partial fraction decomposition of $\frac{1}{z^2 - 3z + 2}$:

$$\frac{1}{z^2 - 3z + 2} = \frac{1}{(z-1)(z-2)} = \frac{A}{z-1} + \frac{B}{z-2} = \frac{(A+B)z + (-2A-B)}{(z-1)(z-2)},$$

and we must have $A+B=0$, $-2A-B=1$. The solution is $A=-1$, $B=1$. Thus,

$$\frac{1}{z^2 - 3z + 1} = -\frac{1}{z-1} + \frac{1}{z-2}.$$

For the integral we have

$$\int_{\gamma} \frac{z^2}{z^2 - 3z + 2} dz = -\int_{\gamma} \frac{z^2}{z-1} dz + \int_{\gamma} \frac{z^2}{z-2} dz = 2\pi i(-1^2 + 2^2) = 6\pi i.$$

Student Number:								
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- (c) The zeros of the denominator $(z^2 + 4)^2$ are $z_{1,2} = \pm 2i$. Notice that $z_1 = -2i$ lies inside γ , and $z_2 = 2i$ lies outside γ .

We use Cauchy's Integral Formula for the first derivative:

$$\int_{\gamma} \frac{e^z}{(z^2 + 4)^2} dz = \int_{\gamma} \frac{e^z}{(z - 2i)^2} \cdot \frac{1}{(z + 2i)^2} dz = \frac{2\pi i}{1!} \left(\frac{e^z}{(z - 2i)^2} \right)' \Big|_{z=-2i}.$$

For the derivative we have

$$\left(\frac{e^z}{(z - 2i)^2} \right)' = \frac{e^z(z - 2i)^2 - e^z 2(z - 2i)}{(z - 2i)^4} = \frac{e^z(z - 2i - 2)}{(z - 2i)^3},$$

$$\left(\frac{e^z}{(z - 2i)^2} \right)' \Big|_{z=-2i} = e^{-2i} \frac{-4i - 2}{(-4i)^3} = e^{-2i} \frac{-4i - 2}{64i} = e^{-2i} \frac{i(4i + 2)}{64} = e^{-2i} \frac{i(2i + 1)}{32}.$$

Thus,

$$\int_{\gamma} \frac{e^z}{(z^2 + 4)^2} dz = 2\pi i \cdot e^{-2i} \frac{i(2i + 1)}{32} = -\pi e^{-2i} \frac{2i + 1}{16}.$$

Student Number:									
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Problem 2

(8 points)

- (a) Formulate the Coincidence Principle for functions f and g that are holomorphic in a region $G \subseteq \mathbb{C}$.
- (b) Does there exist a function f that is holomorphic in the open disc $D(0; 1) = \{z \in \mathbb{C} : |z| < 1\}$ such that

$$f\left(\frac{n-1}{2n}\right) = \frac{3n-1}{2n} \quad \text{and} \quad f\left(\frac{n+1}{2n}\right) = \frac{3n+1}{2n} \quad \text{for all } n \in \mathbb{N}?$$

If yes, how many such functions do exist? Justify your answer.

Solution.

- (a) Let f and g be holomorphic in a region $\mathbb{C}$. If the set $\{z \in G : f(z) = g(z)\}$ has a limit point in G , then $f \equiv g$.
- (b) Both sets $\left\{\frac{n-1}{2n} : n \in \mathbb{N}\right\}$ and $\left\{\frac{n+1}{2n} : n \in \mathbb{N}\right\}$ have a limit point $\frac{1}{2} \in D(0; 1)$.

We have $f\left(\frac{n-1}{2n}\right) = \frac{3n-1}{2n} = 1 + \frac{n-1}{2n}$. By the Coincidence Principle f coincides in $D(0; 1)$ with the function $f(z) = 1 + z$.

Notice that the function $f(z) = 1 + z$ also satisfies $f\left(\frac{n+1}{2n}\right) = 1 + \frac{n+1}{2n} = \frac{3n+1}{2n}$.

Thus, the function $f(z) = 1 + z$ satisfies the desired properties. By the Coincidence Principle there is only one such function.

Student Number:								
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Problem 3

(10 points)

Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be a holomorphic function such that $|f(z)| \leq 1$ for all $z \in \mathbb{C}$ with $|z| = 5$. Show that $|f''(w)| \leq \frac{10}{27}$ for all $w \in \mathbb{C}$ with $|w| = 2$.

Solution.

Consider the circle $\gamma(t) = 5e^{it}$, $t \in [0, 2\pi]$. Let $w \in \mathbb{C}$, $|w| = 2$. By Cauchy's Integral Theorem for derivatives,

$$f''(w) = \frac{2!}{2\pi i} \int_{\gamma} \frac{f(z)}{(z-w)^3} dz.$$

We are going to use the Estimation Theorem. If $|z| = 5$ and $|w| = 2$, then $|z - w| \geq |z| - |w| = 5 - 2 = 3$.

The length of the circle γ is $2\pi \cdot 5 = 10\pi$. We obtain

$$|f''(w)| = \left| \frac{2}{2\pi i} \int_{\gamma} \frac{f(z)}{(z-w)^3} dz \right| \leq \frac{1}{\pi} \frac{1}{3^3} \cdot 10\pi = \frac{10}{27}.$$

Student Number:									
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Problem 4

(12 points)

- (a) Let f be holomorphic in a punctured disc $D'(z_0; r) = \{z \in \mathbb{C} : 0 < |z - z_0| < r\}$ with some $z_0 \in \mathbb{C}$ and $r > 0$. Define the residue $\text{Res}(f, z_0)$.
- (b) Assume that z_0 is a removable singularity of f . What is $\text{Res}(f, z_0)$? Justify your answer.
- (c) For each of the following functions, determine the type of singularity of f at z_0 . If z_0 is a pole, give its order. Calculate the residue $\text{Res}(f, z_0)$.

(i) $f(z) = \frac{\cos \pi z}{z - \frac{1}{2}}, z_0 = \frac{1}{2}$

(ii) $f(z) = \frac{1}{e^z - 1}, z_0 = 0$

Solution.

(a) $\text{Res}(f, z_0) = \frac{1}{2\pi i} \int_{\gamma} f(z) dz$, where $\gamma(t) = z_0 + r_1 e^{it}$, $t \in [0, 2\pi]$, and $0 < r_1 < r$.

- (b) If f has a removable singularity at z_0 , then it can be extended to a holomorphic function on $D(z_0; r)$ by putting $f(z_0) = \lim_{z \rightarrow z_0} f(z)$. Then by Cauchy's Integral Theorem we have $\text{Res}(f, z_0) = \frac{1}{2\pi i} \int_{\gamma} f(z) dz = 0$.

- (c) (i) We have

$$\begin{aligned} \lim_{z \rightarrow \frac{1}{2}} \frac{\cos \pi z}{z - \frac{1}{2}} &= \lim_{z \rightarrow \frac{1}{2}} \frac{\cos \pi z - \overbrace{\cos \frac{\pi}{2}}^{=0}}{z - \frac{1}{2}} = (\cos \pi z)' \Big|_{z=\frac{1}{2}} \\ &= -\pi \sin \pi z \Big|_{z=\frac{1}{2}} = -\pi \sin \frac{\pi}{2} = -\pi. \end{aligned}$$

The limit exists; this means that $z_0 = \frac{1}{2}$ is a removable singularity.

As a consequence, $\text{Res}\left(\frac{\cos \pi z}{z - \frac{1}{2}}, \frac{1}{2}\right) = 0$.

- (ii) For $z \neq 0$

$$f(z) = \frac{1}{e^z - 1} = \frac{\varphi(z)}{z}, \quad \text{where } \varphi(z) = \frac{z}{e^z - 1}.$$

We have

$$\lim_{z \rightarrow 0} \varphi(z) = \lim_{z \rightarrow 0} \frac{z}{e^z - 1} = \lim_{z \rightarrow 0} \frac{z - 0}{e^z - e^0} = \frac{1}{(e^z)'} \Big|_{z=0} = \frac{1}{e^z} \Big|_{z=0} = 1.$$

Student Number:									
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Thus, the function

$$\varphi(z) = \begin{cases} \frac{z}{e^z - 1}, & z \neq 0, \\ 1, & z = 0, \end{cases}$$

is holomorphic in a neighborhood of $z_0 = 0$. Since $\varphi(0) \neq 0$, we conclude that $z_0 = 0$ is a simple pole.

For the residue we have, using the above limit,

$$\text{Res}\left(\frac{1}{e^z - 1}, 0\right) = \lim_{z \rightarrow 0} \frac{z}{e^z - 1} = 1.$$